# Freight Rate Prediction Challenge

## Candidate Technical Report — Muhammad Hanzala Wahid

## 1. Data Quality Findings

The training dataset (train_test.csv) contains 48,000 rows spanning January–October 2025 with 14 columns. The validation set (validation.csv) has 12,000 rows for November–December 2025 with 13 columns (no target). Key data quality issues identified:

| Issue | Column | Train rows | Val rows | Resolution |
|---|---|---|---|---|
| Missing values | weight | 300 (~0.6%) | 165 (~1.4%) | Median imputation (train median) |
| Missing values | market_index | 374 (~0.8%) | 249 (~2.1%) | Median imputation (train median) |
| Unseen cities | pickup/delivery | 64 unique train | 72 unique val | City label enc -> -1 for unseen |
| Target outliers | posted_rate | min=$57, max=$25,538 | N/A | Retained; tree models are robust |

## 2. Validation Strategy

The validation set is a **future time period** (Nov–Dec 2025) relative to the training data (Jan–Oct 2025). A purely random split would cause data leakage through time-correlated features (market_index, quote_signal, temporal patterns). Instead, we use a **time-based expanding-window cross-validation** with 3 folds:

| Fold | Training window | Validation window | Train rows | Val rows |
|---|---|---|---|---|
| 1 | Jan -> Mar 17 (11,998) | Mar 18 -> Jun 1 (12,201) | 11,998 | 12,201 |
| 2 | Jan -> Jun 1 (24,199) | Jun 2 -> Aug 16 (11,968) | 24,199 | 11,968 |
| 3 | Jan -> Aug 16 (36,167) | Aug 17 -> Oct 31 (11,833) | 36,167 | 11,833 |

All preprocessing (median imputation, city label encoding, target encoding) is fitted exclusively on each fold's training portion to prevent any leakage.

## 3. Model Comparison

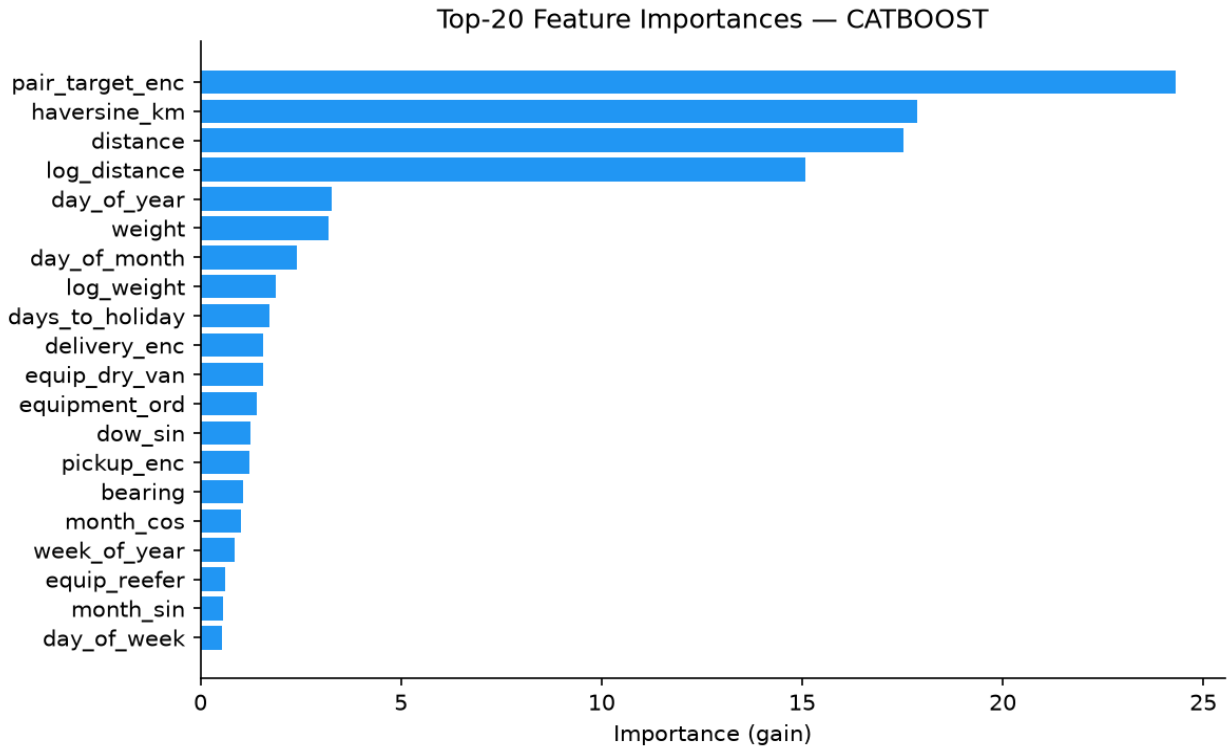
Four models were evaluated using mean metrics across all 3 CV folds. The table below reports MAE, RMSE, MAPE, and R2. Lower MAE/RMSE/MAPE and higher R2 is better.

| Model | MAE ($) | RMSE ($) | MAPE (%) | R2 |
|---|---|---|---|---|
| Median Baseline | 1147.02 | 1567.54 | 68.04 | -0.0732 |
| LightGBM | 199.08 | 674.70 | 9.79 | 0.8013 |
| XGBoost | 204.74 | 679.96 | 10.77 | 0.7981 |
| ★ CatBoost | 175.62 | 645.60 | 8.81 | 0.8180 |

### Model Selection Rationale

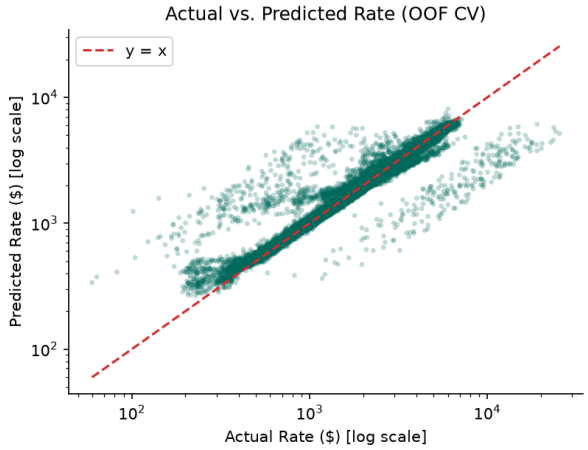
CatBoost had the lowest RMSE on all three folds (mean 645.60 vs LightGBM 674.70, XGBoost 679.96, baseline 1567.54). Optuna hyperparameter tuning (30 trials) was conducted for CatBoost on a 15% temporal holdout to optimize tree depth, learning rate, iterations, and regularization. Note that the CV table uses default params; tuned params are not re-scored in it.

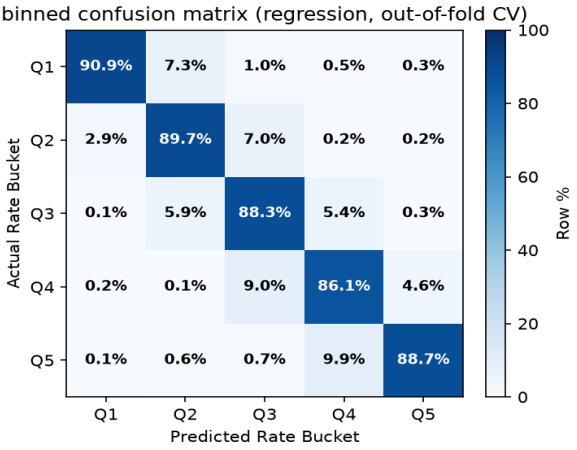
## 4. Feature Importance




Top drivers are pair_target_enc, haversine_km, distance, and log_distance; weight and date features (day_of_year, day_of_month, days_to_holiday) are secondary; quote_signal and market_index are not among the top 20.

## 5. Error Analysis

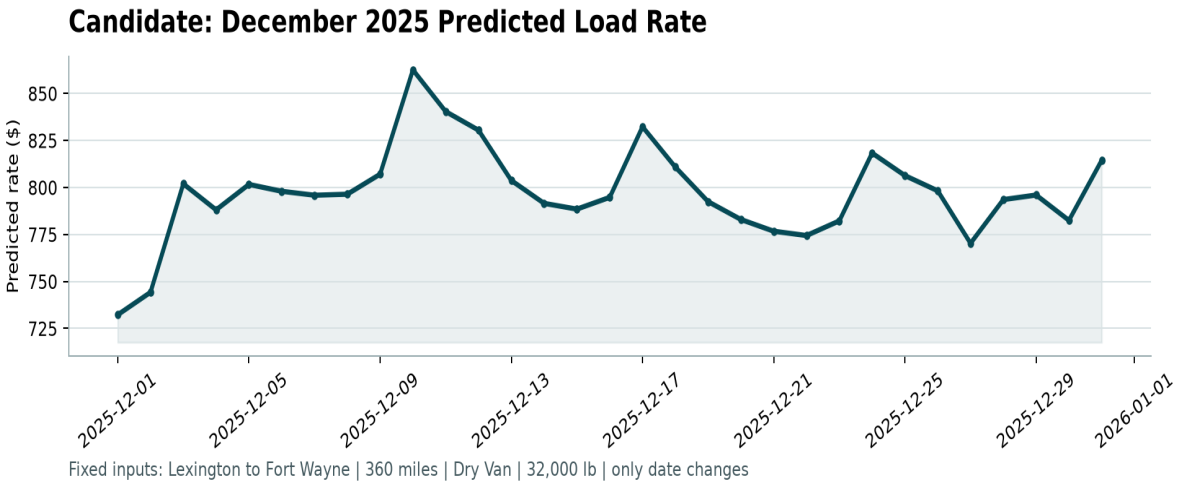



**Actual vs. Predicted Rate (Log-Log)**: The out-of-fold scatter plot demonstrates strong alignment along the ideal y = x reference line across multiple orders of magnitude. The model maintains tight calibration throughout standard freight price regimes with dispersion confined primarily to rare extreme outliers.

**binned confusion matrix (regression, out-of-fold CV)**: Across 5 quantile rate tiers, the model achieves an overall bucket accuracy of 88.75%. Furthermore, 99.14% of predictions fall within at most one adjacent quantile bucket, confirming reliable rate tier ranking without extreme category misclassifications. Note that tuned parameters were selected on the same CV data, so these numbers are evaluation diagnostic estimates rather than final holdout scores.

## 6. December 2025 Predicted Rate Curve




The December curve shows the predicted daily freight rate for the fixed route: Lexington, KY -> Fort Wayne, IN | 360 miles | Dry Van | 32,000 lb. The curve varies through date features only (market_index is set to the training median), and that training covers Jan-Oct only, so this is a plausible estimate, not a validated forecast.

## 7. Limitations

1. In predict.py and features.py, CityCoordLookup builds a lookup of median pickup and delivery coordinates from the 48,000 training rows. Both Lexington and Fort Wayne exist in the training data, so coord_lk.add_coords() maps their exact median coordinates without any hardcoding (unseen cities fall back to the training global median).

2. City-pair target encoding may underperform for rare or unseen routes (72 unique cities in validation vs. 64 in training).

3. All reported metrics are internal CV metrics on a 3-fold time split of train_test.csv. True generalization is assessed by Spotter on the withheld labels.

4. The market_index column has ~0.8% missing values; median imputation may not capture structural breaks in market conditions.

5. Extreme outliers in posted_rate (max $25,533) are retained; clipping could reduce RMSE but may harm calibration on extreme loads.